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EDUCATION

• Indian Institute of Technology Kharagpur

Oct. 2021 – May. 2026 (Expected)

B.Tech.(Hons.) & M.Tech. in Aerospace Engineering; CGPA: 7.94

Kharagpur, India

Minor in Computer Science & Engg. (B.Tech 4Y) | Micro in Artificial Intelligence And Applications

ACHIEVEMENTS: CERTIFICATES

- **2nd Position** Globally among 10k+ participants in PowerPool Alpha Competition by WorldQuant 2025
- **OpenSource Python Package** Published PyPI package with **5.5k+** downloads; pip install **modifiedstemmer**
- **Mentor & Tester** Awarded official recognition by **DeepLearning.AI** for AI For Good Specialization 2023
- **Competitive Programming** Rating **1538** on Codeforces (**abhinav.kgpian**) & **1877** on CodeChef (**abhinav16aero**)

INTERNSHIPS

• Accenture, Noida

Summer Intern

May 2025 – July 2025

- Developed an entity extraction API microservice with multithreading for structured invoice and receipt processing.
- Built multi agent extraction service using **LangGraph** with validator agents, retry policies, pydantic parser.
- Deployed on **GCP Cloud Functions**, with JSON logging and correlation IDs; benchmarked it against DocAI + LLM approach on VertexAI, achieving a avg(throughput) of **2.3 sec/doc**.
- Load tested **100+** documents in Postman, evaluated cost tradeoffs between OpenAI and Gemini, provisioned infrastructure using **Terraform**.

• Business Integration Systems (India) Pvt. Ltd., Pune

GenAI Intern

Feb 2025 – May 2025

- Designed **Rule-based** + **BERT** log analyzer pipeline with regex rules, error validation, PII redaction, and fuzzy confidence scoring, achieving **F1** score of **0.94**.
- Containerized the service using **Docker**, serving Flask API with parallelized batch inference (**12s/100 logs**).
- Added a security layer using MIME type validation and regex filters, with dynamic chunking based log parser.

• Centre for Applied Research in Data Science (CARDS), IIT Ropar

Research Intern

May 2024 – Jul 2024

- Investigated diffusion based (**DDPM**) reinforcement learning methods for trajectory generation and optimization.
- Analyzed classifier guided sampling for planning and proposed a reverse diffusion trajectory reuse technique.
- Trained and evaluated models on NVIDIA **A100** GPUs for the proposed trajectory planning tasks.

• Crafteak, Shivpuri

Data Science Intern

Aug 2023 – Nov 2023

- Developed an advanced recommendation system for personalized food topping suggestions.
- Built a **BERT-NER** + **Word2Vec** pipeline with FAISS based retrieval & rule ranking, improving **NDCG@5** by **4%**.
- Built a **GAN-BERT** model reducing labeled data requirements by **30%** and optimized **1.2 GB** data queries using **PySpark SQL** with partition pruning.

PROJECTS

• Transactional Fraud Detection

Self Project

Mar 2025

- Built an end to end **XGBoost** pipeline for detecting transactional fraud via behavior features, engineered temporal and behavioral features, improving PR AUC score from 0.2129 to 0.9395
- Implemented **class imbalance** handling, threshold tuning, and prevented overfitting by removing dominant features
- Tuned threshold for the leakage safe XGBoost fraud model, achieving **F1 0.9612** on unseen test data

• Predictive Modeling for Kidney Stone Detection | Kaggle Playground Series - S3, E12

Self Project

Apr 2023

- Built a binary classifier for kidney stone prediction, evaluated using **ROC-AUC**, achieving an **AUC of 0.77** and **Rank 39** among **1000+** teams.
- Trained a weighted ensemble of **CatBoost**, **XGBoost**, and **LightGBM**, with ensemble weights tuned using **Optuna** to improve out of fold performance.
- Performed distributional analysis, feature engineering, and statistical **EDA**; applied sampling strategies and used **SHAP** values for model interpretability.

• **Reddit Data Pipeline and T20WC 2024 Data Analysis**

Self Project

Jun 2024 – Jul 2024

- Orchestrated an Apache Airflow pipeline to idempotently ingest **1k+** Reddit posts using Reddit API, transforming data via **Glue ETL** and querying with **Athena**.
- Partitioned data in **Amazon S3** by subreddit keys and developed an end-to-end **RoBERTa** based sentiment analysis pipeline, leveraging **Redshift** for analytics.
- Fetched and processed T20WC 2024 data, loaded it into S3, and analyzed Team India's batting and bowling performance using a **Databricks Spark** cluster.

• **Image Captioning Pipeline | Deep Learning with PyTorch [Course Project], Jovian ML**

Self Project

Mar 2023

- Preprocessed the **Flickr8k** dataset with tokenisation, vocabulary creation, and sequence padding to generate structured caption inputs
- Developed a **CNN-LSTM** model with ResNet50 encoder, LSTM decoder, Bahdanau Attention, with Greedy and Beam Search decoding
- Evaluated caption quality using **BLEU** and **METEOR**, achieving scores of 0.3716 and 0.3795; benchmarked against BLIP VLM and ViT+GPT

• **RAG Engine for Assessment Recommendation**

Self Project

Nov 2025 – Dec 2025

- Built **assessment recommender** by implementing hybrid retrieval with BM25 lexical ranking & dense FAISS vector indexing; used FastAPI.
- Constructed the retrieval indices using **BM25** and **FAISS** embeddings, with TF-IDF for indexing; trained a **learn to rank** reranking mode
- Integrated the generator component in RAG pipeline using **Gemini 2.5 Flash** to produce context grounded outputs using the Gemini API

POSITIONS OF RESPONSIBILITY

• **Student Member, Kharagpur Data Analytics Group**, IIT Kharagpur

Aug 2022 – Apr 2023

- Implemented text to image and style transfer methods for algorithmic art creation using **Stable Diffusion v1.5** for AI art workshops and conducted ML workshops for **100+** participants.
- Organized **KDSH**, India's largest data science event, achieving a **34%** YoY increase in registrations.

COMPETITIONS

• **Ranked 3rd | Marketing Budget Optimization** | General Championships Technology *Feb 2025 – Mar 2025*

- Implemented **Gurobi** based **MISOCP** optimization with risk constraints, achieving **19.2%** higher revenue
- Quantified lag effects using **CatBoost**, validated via **SHAP**, and integrated MMM using the Robyn framework and KPI-KRI evaluation with **HHI** risk analysis to maximize GMV.

• **Ranked 5th | Health Risk Modeling** | General Championships Technology

Feb 2024 – Mar 2024

- Processed SDOH data with **MICE** imputation, used K-Means and BIRCH to classify risk groups (Silhouette=0.62)
- Performed population partitioning and ranked features using Fisher Score and Information Gain; used Gemini API to generate risk cards.

COURSEWORK AND EXTRA CURRICULAR ACTIVITIES : **CERTIFICATES**

- **Courses:** Algorithm (Lab+Theory), AIFA, Graphical models for ML, Interpretable ML, CPS & IOT design, Ubiquitous computing, Embedded system
- **MOOCs:** MLOps, Machine Learning, Deep Learning, Tensor Flow Specialization (Deeplearning.ai), Deep Learning with PyTorch (Jovian), CS229, CS230 (Stanford), Data Analysis with R Programming (Google), Applied Data Science Capstone (IBM), Generative AI (AWS), AlgoZenith
- **Hobbies:** Badminton, Cricket, Multi-player Games, Fine Arts, Learning languages
- **Communication Languages:** English, Hindi, German(Basic)

TECHNICAL SKILLS

- **Languages:** C/C++, Java, Python, R, SQL, MATLAB
 - **Frameworks:** React, Node.js, Django, Express, Tailwind CSS, Flask, FastAPI
 - **Tools & Platforms:** Git, Jupyter Notebook, Google Colab, Docker, AWS, Azure, Hadoop, Linux
 - **Libraries:** NumPy, Pandas, Matplotlib, Keras, OpenCV, Scikit-learn, PySpark, PyTorch
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